Chapter 12

The Ritz Galerkin Method

12.1 Formulation

The theory developed in the previous section can be used also to establish a suitable numerical approximation.

Let $S_h \subset H$ be a *finite-dimensional* subspace of the Hilbert space space H used in Theorem 11.4. Let $a : H \times H \to \mathbb{R}$ be a bilinear form, and let $l : H \to \mathbb{R}$ be a linear functional. We consider the problem of finding $u \in S_h$ such that

$$a(u_h, v_h) = l(v_h), \qquad \forall v_h \in S_h \tag{12.1}$$

Since the finite-dimensional subspace S_h is a Hilbert space in its own right, we can immediately state

Theorem 12.1 *Under the same assumptions as in Theorem 11.4, there is a unique $u \in S_h \subset H$ which solves* (12.1).

Let $n = \dim(S_h)$ be the dimensionality of S_h, and let $(\phi_1, \ldots, \phi_n)$ be a *basis* for S_h. Every $v_h \in S_h$ has a unique representation

$$v_h = \sum_{k=1}^{n} c_k \phi_k. \tag{12.2}$$

Since every $v_h \in S_h$ can be written as a linear combination of the basis function, it is enough to test against all basis functions in (12.1):

$$a(u_h, \phi_j) = l(\phi_j) \qquad j = 1 \ldots, n.$$

Now, using representation (12.2) for u_h, we can write

$$\begin{aligned} a\left(\sum_k c_k \phi_k, \phi_j\right) &= l(\phi_j) \\ \Leftrightarrow \quad \sum_k c_k a(\phi_k, \phi_j) &= l(\phi_j) \qquad j = 1 \ldots, n, \end{aligned}$$

where linearity has been used. Evidently, this is a linear system of equations for the unknown coefficients c_k,

$$A_h \mathbf{z} = \mathbf{b}$$

with

$$A_h = (A_{jk})_{j,k=1,\dots n}, \qquad A_{jk} = a(\phi_k, \phi_j) \tag{12.3}$$

and

$$\begin{aligned} \mathbf{z} &= (c_1, \dots, c_n)^T \\ \mathbf{b} &= (l(\phi_1), \dots, l(\phi_n))^T \end{aligned}$$

Once the approximation space and a basis have been chosen, the matrix A_h and right-hand-side $\mathbf{b}$ can be assembled, and the linear system (12.3) can be solved for the coefficients c_k. These coefficients in turn uniquely determine u_h in the chosen basis (cf. eq. (12.2)).

12.2 An Optimality Statement

We now turn to analyzing the Ritz-Galerkin method. Before quantifying the error we state a best-approximation result, the importance of which can hardly be overestimated:

Theorem 12.2 *Let u and u_h be solutions of* (11.4) *and* (12.1)*, respectively, and let the assumptions of Theorem 11.4 hold. Then*

$$\|u - u_h\|_H \le \frac{C}{\alpha} \inf_{v_h \in S_h} \|u - v_h\|_H ,$$

where C and α are the constants appearing in the Definitions 11.1 and 11.2.

Proof If $u = u_h$, the inequality is trivial. Let in the following $u \neq u_h$. Note that u and u_h satisfy

$$\begin{aligned} a(u, v_h) &= l(v_h) \qquad \forall v_h \in S_h \subset H \\ a(u_h, v_h) &= l(v_h) \qquad \forall v_h \in S_h \end{aligned}$$

and hence

$$a(u - u_h, v_h) = 0, \qquad \forall v_h \in S_h. \tag{12.4}$$

Using both coercivity and continuity of the bilinear form $a(\cdot, \cdot)$ we have

$$\begin{aligned} \alpha \|u - u_h\|_H^2 &\le a(u - u_h, u - u_h) \\ &= a(u - u_h, u - v_h) + \underbrace{a(u - u_h, v_h - u_h)}_{=0}, \qquad \forall v_h \in S_h \\ &\le C \|u - u_h\|_H \|u - v_h\|_H , \qquad \forall v_h \in S_h \end{aligned}$$

Canceling the common factor $\|u - u_h\|$, and noting that $v_h \in S_h$ is arbitrary, we immediately obtain the assertion. ∎

Equation (12.4) is sometimes called *Galerkin orthogonality*. However, only for symmetric bilinear forms is this actually a statement of orthogonality. We come back to this below.

12.3 The Symmetric Case

In this section we investigate a bit more carefully such problems that are characterized by symmetry, i.e. in (11.4) we have

$$a(u, v) = a(v, u), \qquad \forall u, v \in H.$$

(Note that the bilinear form (11.2) corresponding to our Model Problem 4 actually *is* symmetric!) The theorems proved in the previous section apply to both symmetric problems and non-symmetric problems. No new questions regarding well-posedness of the abstract problem arise. However, symmetric problems have some special properties that are often useful and deserve our attention. We assume throughout this section that $a : H \times H \to \mathbb{R}$ is a symmetric bilinear form that satisfies the assumptions of Theorem 11.4.

12.3.1 An Equivalent Norm

The first thing to note is that our bilinear form is now a proper inner product: By assumption a is bilinear and symmetric, and furthermore any coercive bilinear form satisfies

$$a(v,v) \geq 0 \quad \text{and} \quad a(v,v) = 0 \iff v = 0.$$

Then a norm is given by

$$\|v\|_a := \sqrt{a(v,v)}.$$

It is easy to see that this norm is equivalent to the norm $\|\cdot\|_H$, which was used in Definitions 11.1 and 11.2, and on which the assertions of the Lax-Milgram Theorem were built. Indeed, we have, by continuity and coercivity,

$$\alpha \|v\|_H^2 \leq a(v,v) \leq C \|v\|_H^2 \tag{12.5}$$

where α and C are the constants appearing in Definitions 11.1 and 11.2. We had assumed $(H, \|\cdot\|_H)$ is a Hilbert space. Using the equivalence of the norms, it is easy to show that $(H, \|\cdot\|_a)$ is a Hilbert space as well: Firstly, we have for a Cauchy sequence $\{v_n\}_{n\in\mathbb{N}}$ with respect to the norm $\|\cdot\|_a$,

$$\|v_n - v_m\|_H^2 \leq \alpha^{-1} \|v_n - v_m\|_a^2,$$

for any n, m because of coercivity. This means that the sequence is Cauchy also with respect to the norm H. Since H is complete, there is a limit $v \in H$, such that

$$\lim_{n\to\infty} \|v_n - v\|_H = 0.$$

But because of boundedness we have

$$\|v_n - v\|_a^2 \leq C \|v_n - v\|_H^2 \to 0 \quad (n \to \infty),$$

and hence H is complete with respect to the norm $\|\cdot\|_a$.

12.3.2 The Element of Best Approximation

The significance of $(H, \|\cdot\|_a)$ being a proper Hilbert space, it that we can now interpret the solution of the Ritz-Galerkin Method in terms of the following *Best-Approximation Theorem*:

Theorem 12.3 *Let H be a Hilbert space, and S a closed subspace. Then there exists a unique element $u_0 \in S$, such that*

$$\|u - u_0\| \leq \|u - v\|, \quad \forall v \in S, \tag{12.6}$$

where $\|\cdot\| = \sqrt{(\cdot,\cdot)}$, is the norm induced by the scalar product with respect to which the Hilbert space is defined. Furthermore ones has

$$(u - u_0, v) = 0 \quad \forall v \in S \tag{12.7}$$

i.e. $u - u_0$ is orthogonal to all elements $v \in S$.

PROOF First we show that there is an element $u_0 \in S$ that minimizes the distance to u. Obviously the distance $\|u - v\|$ for $v \in S$ is bounded from below (at the very least by zero), so there must be an infimum. We set $d := \inf_{v \in H} \|u - v\|$, and consider a minimizing sequence of functions $\{v_n\}_{n \in \mathbb{N}}$, such that $\|u - v_n\| \to d$. We show that this sequence must be a Cauchy sequence. Since S is closed, the limit function is an element of S (That is to say, the infimum becomes a minimum: There is an element $u_0 \in S$ with $u_0 = \operatorname{argmin}_{v \in S} \|u - v\|$.)

Using the parallelogram law (which can be verified by direct computation)

$$2(x,x) + 2(y,y) = (x - y, x - y) + (x + y, x + y), \quad \forall x, y \in H$$

we have

$$2\|v_n - u\|^2 + 2\|v_m - u\|^2 = \|v_n - v_m\|^2 + 4\left\|\underbrace{\frac{1}{2}(v_n + v_m)}_{\in S} - u\right\|^2$$

Noting $\left\|\frac{1}{2}(v_n + v_m) - u\right\| \geq d$, this can be rearranged to give

$$\|v_n - v_m\|^2 \quad \leq \quad 2\underbrace{\|v_n - u\|^2}_{\to d^2} + 2\underbrace{\|v_m - u\|^2}_{\to d^2} - 4d^2 \to 0 \quad (n, m \to \infty)$$

and hence $\{v_n\}$ is a Cauchy sequence. Since S is a closed subspace, $v_n \to u_0 \in S$, for which by continuity of the norm $\|u - u_0\| = d$.

We now show that this element is orthogonal to all $v \in S$. For arbitrary $v \in S$ and $t \in \mathbb{R}$ there holds

$$\delta(t) := \|u - (u_0 + tv)\|^2 \geq d^2.$$

If u_0 is a minimizer, the function $\delta(t)$ assumes a minimum at $t = 0$, and we must have $\delta'(0) = 0$. Carrying out the differentiation, we obtain

$$\delta'(t) = -2(u - u_0, v) + 2t(v, v)^2,$$

and hence

$$\delta'(0) = 2(u - u_0, v) = 0.$$

Noting that $v \in S$ was arbitrary, we obtain (12.7).

Consider now two elements $u_0, u_1 \in S$, and assume both minimize the distance to u. Then both must fulfill (12.7), and consequently, for arbitrary $v \in S$,

$$(u_0 - u_1, v) = \underbrace{(u_0 - u, v)}_{=0} - \underbrace{(u_1 - u, v)}_{=0} = 0.$$

Choosing $v = u_0 - u_1$, we have

$$\|u_0 - u_1\|^2 = 0,$$

and hence $u_0 = u_1$ almost everywhere. ■

By this theorem, the fact that $(H, \|u\|_a)$ is a Hilbert space, allows us to conclude the following: For any $u \in H$ the element of best approximation, with respect to the norm $\|\cdot\|_a$ in a finite-dimensional (hence closed) subspace $S_h \subset H$, satisfies $a(u - u_h, v_h) = 0$ for all $v_h \in S_h$. In particular, if $u \in H$ solves the abstract problem (11.4), the element of best approximation must then satisfy the Ritz-Galerkin approximation, since

$$0 = a(u - u_h, v_h) = \underbrace{a(u, v_h)}_{=0} - a(u_h, v_h), \qquad \forall v_h \in S_h \subset H.$$

By the best-approximation theorem we also have $\|u - u_h\|_a = \min_{v_h \subset S_h} \|u - v_h\|_a$. We have thus proved:

Theorem 12.4 *Let u and u_h be solutions of* (11.4) *and* (12.1)*, respectively, with a* symmetric *bilinear form. Furthermore, let the assumptions of theorem 11.4 hold. Then*

$$\|u - u_h\|_a = \min_{v_h \subset S_h} \|u - v_h\|_a \,, \tag{12.8}$$

where $\|u\|_a := \sqrt{a(u,u)}$.

12.3.3 The Discrete System

The discrete system arising from the Ritz-Galerkin approximation to the symmetric version of (11.4) also has special properties:

Theorem 12.5 *Let $a : H \times H$ be a symmetric bilinear form that satisfies the assumptions of Theorem 11.4. (Note in particular that this means that a is continuous and coercive!) Then the matrix A_h, defined in* (12.3) *is symmetric positive definite.*

Proof First note that symmetry follows directly from symmetry of the bilinear form $a(\cdot,\cdot)$. Now, let $\mathbf{x} = (x_1, \ldots, x_n)^T \neq \mathbf{0}$ be a vector in $\mathbb{R}^N$. Then we have

$$\begin{aligned}
\mathbf{x}^T A_h \mathbf{x} &= \sum_i \sum_k x_i a_{ik} x_k \\
&= \sum_i x_i \sum_k a(\phi_i, \phi_k) x_k \\
&= a\left(\sum_i x_i \phi_i, \sum_k x_k \phi_k\right) \\
&= a(x_h, x_h) \quad (\text{for some } x_h \in S_h) \\
&\geq \alpha \|x_h\|^2 > 0
\end{aligned}$$

where α is the coercivity constant (cf. Definition 11.2). ∎

12.3.4 A Minimum Principle

For symmetric problems we can connect the variational problem (11.4) to a minimization problem. Let us first formally state this as a theorem:

Theorem 12.6 *Let V be a linear space, and let $a : V \times V \to \mathbb{R}$ be a bilinear form that is*

- *symmetric, i.e. $a(u,v) = a(v,u)$*
- *positive, i.e. $a(u,u) > 0$ for all $0 \neq u \in V$*

Furthermore, let $l : V \to \mathbb{R}$ be a linear functional. Define

$$J(v) := \frac{1}{2} a(v,v) - l(v)$$

Then

$$u = \text{argmin}_{v \in V} J(v) \quad \Longleftrightarrow \quad a(u,v) = l(v) \quad \forall v \in V$$

PROOF We first show the "$\Longleftarrow$" part. Let $u, v \in V$, and $t \in \mathbb{R}$. Assume that $a(u,v) = l(v)$ for all $v \in V$. Then

$$\begin{aligned} J(u+tv) &= \frac{1}{2}a(u+tv, u+tv) - l(u+tv) \\ &= \frac{1}{2}a(u,u) - l(u) + t\underbrace{(a(u,v) - l(v))}_{=0} + \frac{t^2}{2}a(v,v) \qquad (12.9) \\ &= J(u) + \frac{t^2}{2}\underbrace{a(v,v)}_{\geq 0} \\ &\geq J(u) \end{aligned}$$

Now we show the implication "$\Longrightarrow$". let u minimize J. Assume there is $v_0 \in V$ for which $a(u,v) - l(v) = \delta \neq 0$. Then from (12.9):

$$J(u + tv_0) = J(u) + t\delta + \frac{t^2}{2}a(v_0, v_0) \qquad (12.10)$$

Pick $t_0 = -\delta\frac{1}{a(v_0,v_0)}$. (Note that $t_0\delta < 0$, regardless of the sign of δ.) This leads to

$$\begin{aligned} J(u+tv_0) &= J(u) + t_0\delta - \frac{1}{2}t_0\delta \qquad (12.11) \\ &= J(u) + \underbrace{\frac{1}{2}t_0\delta}_{<0} \qquad (12.12) \end{aligned}$$

In contradiction to the assumption. ■

In many cases the minimization of the quadratic functional corresponding to the variational equation has physical meaning. It typically relates to minimization of configurational energy, i.e. the potential energy associated with the various forces within the elements of a physical system.